

Well-Hued Graphs

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Abstract

We define a graph G as well-hued if for each positive integer k there is an integer a_k such that every maximal k -colorable subgraph of G has order a_k . This strengthens the notion of well-covered graphs. In this paper we investigate the connection between this property and related properties, characterize the sequences of the a_k that can be achieved, and determine conditions for the property in some graph classes and under some graph operations such as products.

1 Introduction

We say that an induced k -colorable subgraph H of a graph G is maximal if adding any vertex to H produces a subgraph that is not k -colorable. We define a graph G as **well-hued** if for each positive integer k there is an integer a_k such that every maximal k -colorable subgraph of G has order a_k . An example of a well-hued graph is the 5-cycle, which has $a_1 = 2$, $a_2 = 4$, and $a_k = 5$ for $k \geq 3$.

This notion is rooted in the topic of well-covered graphs. Recall that a graph is **well-covered** if every maximal independent set has the same size. Introduced by Plummer [12], the topic has been investigated in several directions. For example, Ravindra [13] characterized well-covered bipartite graphs, Campbell, Ellingham, and Royle [1] characterized well-covered cubic graphs, Finbow, Hartnell, and Nowakowski [4] characterized well-covered graphs of girth 5 or more, and in a series of papers, Finbow, Hartnell, Nowakowski, and Plummer (see [6]) characterized well-covered planar triangulations.

Several variations of well-covered graphs have been proffered. Most relevant here is that in [8] we defined a graph G to be **well-bicovered** if every maximal bipartite subgraph has the same order. We provided connections and contrasts with well-

covered graphs, and investigated the property and graph operations; in particular, we characterized the maximal outerplanar graphs that are well-bicovered.

In this paper we investigate well-hued graphs. In Section 2, we provide some basic preliminaries comparing with other properties, and characterize the sequences of $\{a_k\}$ that can be realized. In Section 3 we investigate the connection with graph operations, such as join, coronas, and products. In Section 4 we determine the cubic graphs that are well-hued, and in Section 5 we provide results on the planar triangulations that are well-hued. Some further thoughts are given in Section 6.

2 Well- k -Partite Graphs and Well-Hued Sequences

We begin by observing that the property well-hued is distinct from well-covered and well-bicovered and their generalizations. Then we characterize the $\{a_k\}$ sequences that can be achieved.

2.1 Well- k -partite graphs

For a graph G we denote by $\alpha_k(G)$ the maximum order of a k -partite subgraph. We define a graph G as **well- k -partite** if every maximal k -partite subgraph has the same order, namely $\alpha_k(G)$. In this terminology, well-1-partite is the same as well-covered, and well-2-partite the same as well-bicovered.

We note that these are independent concepts. For example, for all $\ell \geq 1$ one can construct graphs that are well- j -partite for $1 \leq j \leq \ell$ but not well- $(\ell + 1)$ -partite. One construction of such a graph S_ℓ is to take a clique A of cardinality $\ell + 1$ and a clique B of cardinality $\ell + 2$ and choose vertices x of A and y of B and join x to all vertices of B except y . The graph S_2 is illustrated in Figure 1. Every maximal j -partite set of S_ℓ has cardinality $2j$ for $1 \leq j \leq \ell$, but omitting either $\{x, y\}$ or one vertex of $B - \{y\}$ yields a maximal $(\ell + 1)$ -partite subgraph.

Further, it is easy to create an ℓ -colorable graph T_ℓ that is not well- j -partite for any $j < \ell$ (for example, $K_{\ell+1} - e$). By using the graphs S_ℓ and T_ℓ as components, one can construct for any k and subset J of $\{1, \dots, k\}$ a graph that for $j \in \{1, \dots, k\}$ is well- j -partite for precisely $j \in J$.

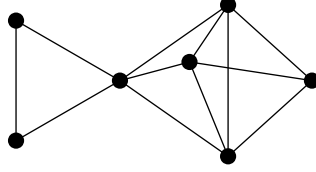


Figure 1: The graph S_2 that is well-1- and well-2- but not well-3-partite

2.2 Well-Colored Graphs and Super Well-Covered Graphs

We introduced well-hued as a generalization of well-covered. Another way to generalize the concept of a graph being well-covered is to consider a coloring process. Define Process C as follows:

PROCESS C. Use the first color as much as possible. Then the second color as much as possible and so on. Equivalently, color 1 is any maximal independent set, color 2 is any maximal independent set of what remains, and so on.

The maximum number of colors used in Process C is the **Grundy number** of the graph (and a Grundy coloring of the vertices with positive integers is one where every vertex has neighbors of all smaller colors). Zaker [16] defined a graph to be **well-colored** if the Grundy number and chromatic number are equal. A stronger notion than this is the requirement that for each positive integer k the Process C colors the same number of vertices with color k . Fradkin [7] called such a graph **super well-covered**.

It is easy to see that being well-hued does not imply the graph is super well-covered or even well-colored: the path P_4 is an example. Similarly, there are many graphs that are well-colored but not well-hued: the path P_3 is an example. On the other hand, being super well-covered is a stronger condition than being well-hued:

Lemma 1 *If a graph G is super well-covered, then it is well-hued.*

Proof. We need to show that the graph G is well- k -partite for every k . Consider a maximal k -partite subgraph H and a color class C_1 of a k -coloring of H . Then one can extend C_1 to a maximal independent set C_1^* of G ; necessarily all added vertices are already in H , since otherwise we could add them to the colored H and give them

the color of C_1 , contradicting the maximality of H . Similarly, if we let C_2 be a color class of a $(k-1)$ -coloring of $H - C_1^*$, this can be extended to a maximal independent set C_2^* of $G - C_1^*$. That is, there is an execution of Process C such that a vertex is in H if and only if it receives a color in the range 1 up to k . Since G is super well-covered, all such H have the same order. QED

2.3 Well-hued sequences

The following lemma has probably been noticed before. (Note though that the greedy approach of letting X_1 be any maximal independent set fails: for example, for $G = P_4$ and taking the two leaves.) For example, Catlin [2] showed a minimum coloring can be chosen where one color class is a maximum (not just a maximal) independent set.

Lemma 2 *If a graph G is m -colorable, then there exists a proper coloring with color classes $X_1, X_2, \dots, X_m$ such that for $1 \leq k \leq m$ the set $\bigcup_{i=1}^k X_i$ induces a maximal k -colorable subgraph of G .*

Proof. Build a series of subgraphs A_k as follows. Let A_m be the whole graph. Then for $k = m-1$ down to 1 do: let A_k be a maximal k -colorable subgraph of A_{k+1} subject to the constraint that $X_{k+1} = V(A_{k+1}) - V(A_k)$ is an independent set. Finally let $X_1 = V(A_1)$.

We claim that each A_k is a maximal k -colorable subgraph in the whole graph G . The proof is by induction on decreasing k . The base case is true for $k = m$.

Now suppose some vertex v of the graph can be added to A_k and one still has a k -colorable subgraph. If v is in A_{k+1} , then we have a contradiction of the choice of A_k (as the rest of A_{k+1} remains an independent set). If v is outside A_{k+1} , then the set $(V(A_k) \cup \{v\}) \cup X_{k+1}$ induces a $(k+1)$ -colorable subgraph containing A_{k+1} , a contradiction. QED

We are now in a position to characterize the $\{a_k\}$ sequences achieved by well-hued graphs.

Theorem 1 *There is a well-hued graph with sequence $\{a_k\}_{k \geq 1}$ if and only if the sequence $\{a_k - a_{k-1}\}_{k \geq 1}$ (where we define $a_0 = 0$) is non-increasing and eventually zero.*

Proof. Assume graph G is well-hued. Consider the coloring of G provided in Lemma 2. For each k and for every $\ell \geq k$, the set $X_\ell \cup \bigcup_{i=1}^{k-1} X_i$ induces a k -colorable subgraph. By the well-huedness it cannot have size bigger than a_k . Thus $a_{\ell+1} - a_\ell = |X_\ell| \leq |X_k| = a_k - a_{k-1}$. Also, a_k is the order of the graph for all $k \geq \chi(G)$.

To show that the sequence $\{a_k\}_{k \geq 1}$ is achievable, we construct a graph G that is a disjoint union of cliques. Let $d_1 = a_1$ and $d_i = a_i - a_{i-1}$ for $i \geq 2$. For each $i \geq 1$, let G contain $c_i = d_i - d_{i+1}$ copies of K_i . Since the values d_i are non-increasing, all the c_i are nonnegative; since the differences become zero, this produces a finite graph. Further, every maximal k -partite subgraph of the resultant graph has order

$$\begin{aligned}
& \sum_{i=1}^{k-1} i c_i + \sum_{i=k}^{\infty} k c_k \\
&= (d_1 - d_2) + 2(d_2 - d_3) + 3(d_3 - d_4) + \dots + (k-1)(d_{k-1} - d_k) + k \sum_{i=k}^{\infty} (d_i - d_{i+1}) \\
&= d_1 + d_2 + d_3 + \dots + d_k \\
&= a_k,
\end{aligned}$$

as required. QED

Apart from which sequences are attainable, one can also ask how often a particular sequence is attained. For example, consider just the values a_1 and a_2 . We observed that $a_2 \leq 2a_1$. Well-hued graphs with $a_2 = 2a_1$ are plentiful; but graphs where a_2 is close to a_1 seem sparser. For example:

Lemma 3 *Let G be a well-hued graph without isolated vertices and with $a_2 = a_1 + 1$. Then G is a complete graph.*

Proof. Consider a well-hued graph G with $a_2 = a_1 + 1$. Consider any maximal independent set A of G and let $B = V(G) - A$. Now extend A to a maximal bipartite subgraph of G . The restriction on a_2 implies that we are only able to add one vertex from B . It follows that B is a clique.

Now, create a maximal independent set A' of the graph G by starting with a vertex b of B . Since $|A'| = |A|$ we have that A' contains all but one vertex of A . Therefore there is exactly one vertex in A adjacent to b ; call it v_b .

Now, create a maximal bipartite subgraph H of G by starting with two vertices b_1 and b_2 of B . (If B contains only one vertex then the graph G is a star and so

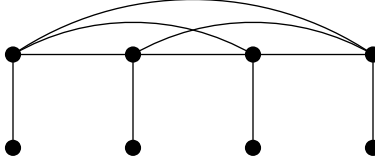


Figure 2: A well-hued graph with $a_2 = a_1 + 2$

must be K_2 .) Since graph G is well-bicovered, the graph H must contain all but one vertex of A . Thus vertices b_1 and b_2 have exactly one common neighbor in A ; that is, $v_{b_1} = v_{b_2}$. It follows that there is one vertex in A adjacent to all of B , which creates a clique with $|B| + 1$ vertices. There cannot be any other vertex in A , since it would have no neighbor in B and hence be isolated. QED

Consider next the graphs with $a_2 - a_1 = 2$. There are many graphs with $a_1 = 2$ and $a_2 = 4$, or with $a_1 = 3$ and $a_2 = 5$. However, computer search suggests that:

Conjecture 1 *For $a_1 \geq 4$, the only connected well-hued graph with $a_2 = a_1 + 2$ is the corona of the complete graph, an example of which is shown in Figure 2.*

Indeed it might be true that for fixed c the well-hued graphs with $a_2 \leq a_1 + c$ are polynomial-time recognizable.

3 Graph Operations

In this section we consider the relationship between the property of being well-hued and a few graph operations and related graph families.

3.1 Union, Join, and Corona

It is trivial to see that the disjoint union of two well-hued graphs is well-hued. Indeed one can even add a bridge to the disjoint union joining the two components, provided both graphs are nontrivial.

We consider next the join of two graphs (obtained by taking their disjoint union and adding all edges between them). We say a k -chromatic graph G of order n is **well-equi-hued** if it is well-hued with $a_i = \frac{i}{k}n$ for $i \in \{1, \dots, k\}$. For example, the octahedron $K_{2,2,2}$ is well-equi-hued.

Theorem 2 *Let G and H be graphs and let J be their join. The join J is well-hued if and only if $\alpha_1(G) = \alpha_1(H)$ and both G and H are well-equi-hued.*

Proof. Suppose first that J is well-hued and let $A \cup B$ be the vertices of a maximal k -colorable subgraph of J where $A \subseteq V(G)$ and $B \subseteq V(H)$. Assume the subgraph induced by A has chromatic number ℓ (where the graph induced by the empty set has chromatic number 0). Then A must be maximal ℓ -partite and B must be maximal $(k - \ell)$ -partite. In particular, by considering all possible J it follows that both G and H are well-hued.

We will use primes to indicate the parameters for H . Let us assume that $\chi(G) = r$. For $k = 1$ it is required that $a_1 = a'_1$. That is, G and H have the same independence number. For $k = 2$ it is necessary that $a_2 = a_1 + a'_1$. Since $a_1 = a'_1$, this is equivalent to the condition that $a_2 = 2a_1$. Thus, continuing this argument, for all $2 \leq i \leq r$ it must be true that $a_i = a_{i-1} + a'_1 = (i - 1)a_1 + a_1 = ia_1$. On the other hand, we know that $a_r = |V(G)|$. Therefore $a_1 = |V(G)|/r$, which implies $a_i = ia_1 = i|V(G)|/r$ and G is well-equi-hued. Reversing the roles of a_i and a'_i shows that H is also well-equi-hued.

Conversely, if G and H have the stated properties, then one can readily show that every maximal k -partite subgraph of J has order $k\alpha_1(G)$ for each k up to the chromatic number of J (the sum of the chromatic numbers of G and H). QED

3.2 Coronas and Split Graphs

Recall that the **corona** of a graph G , which we denote $Cor(G)$, is obtained from G by introducing for each vertex v of G a new vertex v' adjacent only to v . It is easy to see that for $k \geq 2$ every maximal k -partite subgraph of $Cor(G)$ consists of a maximal k -partite subgraph of G together with all the new vertices. On the other hand, it is easy to see (and known) that $Cor(G)$ is always well-covered. In particular, one obtains the following theorem:

Theorem 3 *A graph G is well-hued if and only if G is well-covered and $Cor(G)$ is well-hued.*

Recall that a **split graph** is one whose vertex set has a partition into a clique and an independent set.

Theorem 4 *If G is a connected, well-hued split graph, then either $G = K_n$ or $G = \text{Cor}(K_n)$.*

Proof. Assume G has partition (C, I) such that I is a maximum independent set. Suppose some vertex $u \in C$ has two neighbors in I , say x_1 and x_2 . Then $\{u\} \cup (I - \{x_1, x_2\})$ is maximum independent, implying that G is not well-covered. Thus, every vertex of C has at most one neighbor in I . However, if some vertex of C has no neighbor in I , then I is not maximum independent. Thus, each vertex of C has exactly one neighbor in I . Next, suppose some $v \in I$ has two neighbors in C , say y_1 and y_2 . If v is adjacent to every vertex of C , then $G = K_n$ and we are done. So we shall assume that v is not adjacent to $w \in C$. Then $\{y_1, w\} \cup I$ and $\{y_1, y_2\} \cup (I - \{v\})$ are both maximal bipartite implying G is not well-bicovered. Therefore, $G = \text{Cor}(K_n)$. QED

3.3 Lexicographic product

We consider the lexicographic product, also known as composition. (Recall that the lexicographic product $G[H]$ of graphs G and H has vertex set $V(G) \times V(H)$ and vertices (g_1, h_1) and (g_2, h_2) are adjacent if $g_1 g_2 \in E(G)$ or $g_1 = g_2$ and $h_1 h_2 \in E(H)$.) Topp and Volkmann [15] showed that if both graphs have at least one edge, then the lexicographic product of graph G with graph H is well-covered if and only if both G and H are well-covered.

In [8] we showed that for the product to be well-bicovered, one needs G only to be well-covered but needs H to be more than well-bicovered. The following theorem generalizes that result:

Theorem 5 *Assume graph G has at least one edge and graph H has chromatic number at least k . Let $L = G[H]$ denote the lexicographic product of G with H . Then the product L is well- k -partite if and only if*

- (i) G is well-covered; and
- (ii) $\alpha_k(H) = k\alpha_1(H)$, and H is well- i -partite for $1 \leq i \leq k$.

Note that condition (ii) implies that $\alpha_i(H) = i\alpha_1(H)$ for $1 \leq i \leq k$.

Proof. Assume L is well- k -partite. Let J be a maximal independent set of G and K be the vertices of a maximal k -partite subgraph of H and let $S = J \times K$. Since H

has chromatic number at least k , the set S induces a maximal k -partite subgraph of L . (Any vertex outside S has neighbors of all k colors.) So that all possible S have the same order, all possible J must have the same order; that is, G is well-covered. Similarly we need H to be well- k -partite.

Now, fix some edge uv of G . Then there exists a maximal independent set J_u of G that contains u but not v , and a maximal independent set J_v of G that contains v but not u . For each $i \in \{1, \dots, k\}$ let K_i be the vertex set of some maximal i -partite subgraph of H , and let K_0 be the empty set. Then for each $j \in \{0, 1, \dots, k-1\}$ define the set T_j of vertices in the product by

$$T_j = ((J_u - J_v) \times K_j) \cup ((J_v - J_u) \times K_{k-j}) \cup ((J_u \cap J_v) \times K_k).$$

A proper k -coloring of the subgraph induced by T_j is obtained by using j colors on $(J_u - J_v) \times K_j$, then $k - j$ different colors on $(J_v - J_u) \times K_{k-j}$, and then all k colors on $(J_u \cap J_v) \times K_k$. Further, we claim that the subgraph T_j is maximal. For, suppose one can add some vertex to T_j and the result is still k -colorable. Then let $c(i)$ be the number of copies of H in which color i is used. It follows that $\sum_i c(i) > j|J_u - J_v| + (k - j)|J_v - J_u| + k|J_u \cap J_v| = k\alpha_1(G)$. This means that some color is used on more than $\alpha_1(G)$ copies of H , a contradiction.

To get the same number of vertices in T_j in each case, it is necessary that H is well- j -partite. Furthermore, to get that $|T_j|$ is the same for all j , it must be that $\alpha_j(H) = j\alpha_1(H)$.

On the other hand, assume the conditions on G and H are satisfied. Consider a maximal k -partite subgraph T of the product L with some k -coloring. Consider say the red vertices. Their projection onto G must form an independent set. If this projection is not a maximal independent set, then there is some vertex of G , say x , that is not adjacent to any vertex in the projection. This means that some vertex of the copy H_x can be added to T and given color red, a contradiction. That is, each color projects to a maximal independent set of G and thus is used in $\alpha_1(G)$ copies of H . Furthermore, by hypothesis, each color used in a particular copy H_v of H is used $\alpha_1(H)$ times in H_v . Thus $|T| = k\alpha_1(G)\alpha_1(H)$ and so L is well- k -partite. QED

The question when H has chromatic number less than k seems more complicated. This is inescapable here, since requiring the product to be well-hued always requires considering more colors than the chromatic number of H . A small computer search suggests that the following might be true:

Question 1 *Is it true that the lexicographic product $L = G[K_2]$ of graph G with K_2 is well-3-partite if and only if G is both well-covered and well-bicovered and $\alpha_2(G) = 2\alpha_1(G)$?*

Indeed, the computer search suggests that $G[K_2]$ is well-hued under the same conditions. It is not hard to see that there is some connection between independent sets and bipartite subgraphs of G on the one hand and 3-colorable subgraphs of L on the other. Nevertheless, we are unable to prove that the conditions are either sufficient or necessary. It is also possible that fractional colorings of G play a role. It is not necessary that G be well-hued. For example, consider the graph S_2 from Figure 1: (a computer calculation shows that) the lexicographic product $S_2[K_2]$ is well-hued even though S_2 is not.

3.4 Direct products

We look next at the direct product sometimes called the tensor product. (Recall that the direct product $G \times H$ of graphs G and H has vertex set $V(G) \times V(H)$ and vertices (g_1, h_1) and (g_2, h_2) are adjacent if $g_1g_2 \in E(G)$ and $h_1h_2 \in E(H)$.)

Topp and Volkmann established the following necessary condition for the direct product to be well-covered.

Theorem 6 [15] *Let G and H be graphs without isolated vertices. If $G \times H$ is well-covered, then both G and H are well-covered and $\alpha_1(G)|V(H)| = \alpha_1(H)|V(G)|$.*

Topp and Volkmann also provided a characterization where one of the factors is very well-covered. Recall that a graph is **very well-covered** if it is well-covered and $\alpha_1(G) = |V(G)|/2$.

Theorem 7 [15] *Let G and H be graphs without isolated vertices. If at least one of G and H is very well-covered, then the product $G \times H$ is well-covered if and only if both G and H are very well-covered.*

Examples of direct products that are well-hued are readily available where one of the factors is bipartite. This is because the resultant product is bipartite, and so it is sufficient that the product be well-covered. Such a case includes the direct product of two coronas, at least one of which is bipartite.

However, we do not know of an example where the product is well-hued and neither factor is bipartite. An obvious candidate is $K_n \times K_n$, since Topp and Volkmann [15] noted that the product is well-covered. However:

Lemma 4 *For $n \geq 3$ the product $K_n \times K_n$ is not well-bicovered.*

Proof. Let $V(K_n \times K_n) = \{(i, j) : 0 \leq i, j \leq n-1\}$. Then the subgraph of all vertices of the form $(i, *)$ with $i \in \{0, 1\}$ is maximal bipartite and has order $2n$. But the subgraph of all vertices of the form $(0, *)$ and $(*, 0)$ is also maximal bipartite and has order $2n - 1$. QED

One can also consider products where only one factor is complete. (Indeed, in [11] it was shown that if the direct product is well-covered then each factor G is either complete or has a vertex x and an independent set I such that x is isolated in $G - N[I]$.) Here is a necessary condition for the result to be well-bicovered.

Theorem 8 *Let G be a graph of order n . If $G \times K_m$ for $m \geq 3$ is well-bicovered, then G is well-bicovered.*

Proof. Note that if G is bipartite, then it is well-bicovered. So assume that G is not bipartite. Let $V(K_m) = \{0, \dots, m-1\}$. Pick a maximal bipartite subgraph G' of G . We claim that $G' \times K_m$ is maximal bipartite in $G \times K_m$. It is certainly bipartite. Consider any other vertex of the product: say $(v_1, 0)$ where $v_1 \in V(G) - V(G')$. Since the subgraph G' is maximal bipartite, there exist vertices $v_2, \dots, v_c$ in G' such that $v_1 v_2 \dots v_c$ is an odd cycle in G . Consider the graph induced by $V(G' \times K_m)$ together with $(v_1, 0)$. Then

$$\{(v_1, 0)\} \cup \{(v_{2k}, 1) : 1 \leq k \leq \lfloor c/2 \rfloor\} \cup \{(v_{2k+1}, 2) : 1 \leq k \leq \lfloor c/2 \rfloor\}$$

induces an odd cycle in $G \times K_m$. Thus, $G' \times K_m$ is maximal bipartite. Note that this is true for any choice of G' . Thus, every maximal bipartite subgraph of G has the same cardinality. QED

Indeed, we know of only one example of a well-bicovered product with neither factor being bipartite. This is the direct product $F \times K_3$, where F is K_4 with a foot at two different vertices, shown in Figure 3.

Moreover, we do not know of any example where $G \times K_3$ is well-hued (and G has at least one edge). We know there is no small example, as we show next:

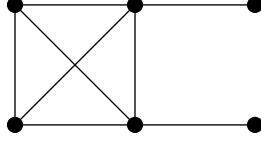


Figure 3: The graph F such that $F \times K_3$ is well-bicovered

Theorem 9 *If G is a connected graph with at least one edge and $G \times K_3$ is well-hued, then $|V(G)| \geq 9$.*

We will use the following result:

Lemma 5 ([11]) *Let $n \geq 2$. If $G \times K_n$ is well-covered, then for every $z \in V(G)$ such that $\deg(z) \geq n$, the graph $G - N[z]$ has at least one isolated vertex.*

Proof of Theorem 9. Let G have order $n \leq 8$ and suppose $G \times K_3$ is well-hued. By Theorem 6 it follows that G is well-covered. Let $V(K_3) = \{0, 1, 2\}$. If J is any maximal independent set of G , then $J \times \{0, 1, 2\}$ is a maximal independent set in the product; at the same time so is the set $V(G) \times \{0\}$. It follows that $3\alpha_1(G) = n$. In particular, n is a multiple of 3. If $n = 3$ then $G = K_3$, but by Lemma 4 we know that $K_3 \times K_3$ is not well-bicovered. It follows that $n = 6$. Now, $V(G) \times \{0, 1\}$ is maximal bipartite, as is $V(G') \times \{0, 1, 2\}$ for any maximal bipartite subgraph G' of G . Thus, all maximal bipartite subgraphs of $G \times K_3$ have order 12 and all maximal bipartite subgraphs of G have order 4.

In particular, it follows that G is not bipartite. One can easily check by the characterization of [4] that G cannot have girth 5. Therefore, it contains a triangle.

Claim If z is a vertex of a triangle T in G with degree at least 3, then z has degree 4 and the two vertices in $N[z] - V(T)$ are not adjacent.

Proof of Claim: By Lemma 5, $G - N[z]$ contains an isolated vertex, say u . Since G is well-covered with $\alpha_1(G) = 2$, $\{z, u\}$ is a maximal independent set. Thus z is adjacent to all vertices of G except u ; that is, it has degree 4. If the two vertices in $N[z] - V(T)$ are adjacent, then $(V(G) \times \{0\}) \cup (\{u, z\} \times \{1, 2\})$ is maximal bipartite of order 10, which is a contradiction. (qed)

Now, let $T = \{x, y, z\}$ be a triangle in G , and let $N[z] - V(T) = \{v, w\}$. Since G is well-covered, the vertex x must be adjacent to at least one of v or w , say v . Then

by the Claim applied to z as part of triangle $T' = \{x, v, z\}$, it follows that y and w are not adjacent. This forces $vy \in E(G)$.

Now, applying the Claim to vertex y in triangle T , we conclude that yu is an edge in $E(G)$ and that vu is not. But by the Claim applied to vertex v in triangle T' , it follows that vu is an edge in $E(G)$ and that yu is not. A contradiction. QED

3.5 Cartesian Products

In our investigation of the Cartesian product we start with the Cartesian product of two complete graphs, sometimes called a rooks graph. (Recall that the Cartesian product $G \square H$ of graphs G and H has vertex set $V(G) \times V(H)$ and vertices (g_1, h_1) and (g_2, h_2) are adjacent if $g_1 = g_2$ and $h_1 h_2 \in E(H)$ or $h_1 = h_2$ and $g_1 g_2 \in E(G)$.)

Lemma 6 *Let $n \geq m$ be positive integers and let $R = K_m \square K_n$. Then for $k \geq 1$, the graph R is well- k -partite if and only if $k \geq m$ or $k \leq n - m + 1$.*

Proof. Recall that R is perfect (it is the line graph of $K_{m,n}$). That is, for every subgraph the chromatic number equals the clique number.

Assume first that $k \geq m$ and let G be a maximal k -partite subgraph of R . Then G must contain $\min(k, n)$ vertices in every copy of K_n , since otherwise one can just add any vertex in such a K_n that does not contain k vertices of G . It follows that G has order $\min(km, nm)$.

Assume second that $k \leq n - m + 1$ and let G' be a k -partite subgraph of R with order less than km . Then there must exist a copy of K_n , call it H , that contains at most $k - 1$ vertices of G' . Further, there can be at most $m - 1$ copies of K_m that contain k vertices. Thus there must be at least $n - (k - 1 + m - 1)$ vertices in H that we can add to G' while maintaining k colorability. When $k \leq |n - m| + 1$, it holds that $n - (k - 1 + m - 1) \geq 1$ and so G' is not maximal. That is, all maximal k -partite subgraphs have order km .

Assume third that $n - m + 1 < k < m$. Then one can easily find an k -partite subgraph of R with order km . We will now find one with smaller order. Denote the vertices of R by $\{(i, j) : i \in \{1, \dots, m\}, j \in \{1, \dots, n\}\}$ and let J be the subgraph of R induced by $\{(i, j) : i \in \{1, \dots, m - 1\}, j \in \{1, \dots, m - 1\}\}$. Let J' denote an k -colorable subgraph of J with k vertices from each copy of K_{m-1} . Let D be the

vertices $\{(m, j) : j \in \{m, \dots, n\}\}$. Note that $|D| = n - m + 1 < k$. Let K be the subgraph of R induced by $D \cup V(J')$. Then K is k -colorable (there are no edges between D and J') and K has order at most $km - 1$. Thus R is not well- k -partite. QED

We obtain the following corollary:

Theorem 10 *For $n \geq m$, the product $R = K_m \square K_n$ is well-hued if and only if $n \geq 2(m - 1)$.*

Proof. If $m \leq n < 2(m - 1)$, then by the above lemma R is not well- $(m - 1)$ -partite, since $n - m + 1 < (2(m - 1) - m) + 1 = m - 1$. On the other hand, if $n \geq 2(m - 1)$ then $n - m + 1 \geq (2(m - 1) - m) + 1 = m - 1$; thus there exists no integer k such that $n - m + 1 < k < m$. It follows that R is well-hued. QED

Well-hued products where one of the factors is complete seem common: for example, the product $B \square K_2$ where B is the graph from Figure 4 is well-hued. But there are well-hued products where neither factor is complete: for example, the product $C_5 \square H$ where H is K_5 with one foot (according to computer check).

In [9], Hartnell and Rall showed that if $G \square H$ is well-covered, then one of the factors must be well-covered. We do not know the answer to the analogous question for well-hued.

We conclude this section with the case that both factors have girth 5.

Theorem 11 [10] *Let G and H be nontrivial, connected graphs, both of which have girth at least 5. The product $G \square H$ is well-covered if and only if $G \square H \cong K_2 \square K_2$ or $G \square H \cong C_5 \square K_2$.*

It can easily be checked that these two products are well-hued.

4 Cubic Graphs

Campbell et al. [1] characterized the well-covered cubic graphs. The characterization is a class of graphs constructed from the “graph fragments” A , B , and C shown in Figure 4. For a fragment, define a *terminal pair* to be a pair of adjacent degree-two vertices (shown in white). Let $\mathcal{W}$ denote the set of cubic graphs constructed as

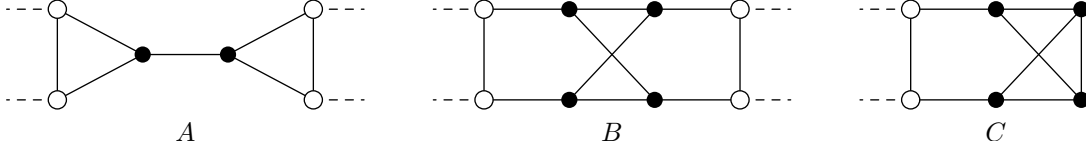


Figure 4: Graph “fragments” A , B , and C

follows. Given a collection of copies of A , B , and C , join every terminal pair by two edges to another terminal pair so that the result is cubic.

Let $K_{3,3}^*$ be the graph obtained from $K_{3,3}$ by replacing one vertex with a triangle and let E_{12} be the graph obtained from the 3-cube by replacing two antipodal vertices with a triangle. Let P_{14} be the generalized Petersen graph shown in Figure 5.

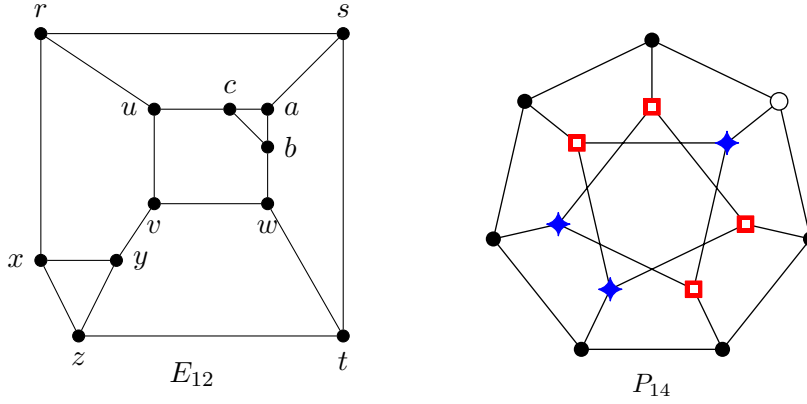


Figure 5: Cubic well-covered graphs

Theorem 12 [1] *Let G be a connected cubic graph. Then G is well-covered if and only if one of the following is true:*

- (i) $G \in \mathcal{W}$; or
- (ii) G is one of the six exceptional graphs: K_4 , $K_{3,3}$, $K_{3,3}^*$, $C_5 \square K_2$, E_{12} , or P_{14} .

Lemma 7 *Neither $K_{3,3}^*$ nor P_{14} is well-bicovered.*

Proof. To see that $K_{3,3}^*$ is not well-bicovered, let xyz represent the triangle and B represent the vertices that are adjacent to exactly one vertex of the triangle. Then $G - \{x, y\}$ is maximal bipartite and so is $B \cup \{x, y\}$.

To see that P_{14} is not well-bicovered, note that the subgraph induced by the black vertices and the red (hollow square) vertices is maximal bipartite, as is the subgraph induced by the black vertices and the blue (star) vertices, but these subgraphs have different orders. QED

Lemma 8 E_{12} is well-bicovered.

Proof. We show that every maximal bipartite subgraph has order 8. Note that E_{12} is obtained from the 3-cube by a Y - Δ operation at an antipodal pair of vertices. Thus all six vertices in triangles are in the same orbit under the automorphism group.

Let M be any maximal bipartite subgraph and label the vertices of E_{12} as shown in Figure 5. Note that $|V(M) \cap \{a, b, c\}| \leq 2$ and $|V(M) \cap \{x, y, z\}| \leq 2$. Suppose first that $|V(M) \cap \{a, b, c\}| = 2$. By symmetry, we may assume that $V(M) \cap \{a, b, c\} = \{b, c\}$. Thus, at least one of u, v , or w is not in M . Suppose first that $u \notin M$. Thus, one of v, w, t, y , or z is not in M . If $y \notin V(M)$, then one of r, s, t, z , or x is not in M . One can easily verify that $V(E_{12}) - \{a, u, y, \ell_y\}$ where $\ell_y \in \{r, s, t, z, x\}$ is bipartite. If $z \notin V(M)$, then one of r, s, t, w, v, x , or y is not in M . Again one can verify that $V(E_{12}) - \{a, u, z, \ell_z\}$ where $\ell_z \in \{r, s, t, w, v, x, y\}$ is bipartite. Suppose next that $v \notin V(M)$. Then one of x, y , or z is not in M . Note that $V(E_{12}) - \{a, u, v, \ell_v\}$ where $\ell_v \in \{x, z\}$ is bipartite. However, if $y \notin V(M)$, then M is not maximal as we could have included v . Next, suppose $w \notin V(M)$. Then one of r, s, t, x , or z is not in M . Note that $V(E_{12}) - \{a, u, w, \ell_w\}$ where $\ell_w \in \{x, z\}$ is bipartite. However, if $y \notin V(M)$, then M is not maximal as we could have included w . Finally, assume $t \notin V(M)$. Then $V(E_{12}) - \{a, u, t, \ell_t\}$ where $\ell_t \in \{x, y, z\}$ is bipartite.

The case that $w \notin V(M)$ is symmetric to the case $u \notin V(M)$. So we now assume instead that $v \notin V(M)$ and $\{b, c, u, w\} \subset V(M)$. Thus, at least one of $\{x, y, z\}$ and at least one of $\{r, s, t\}$ is not in M . On the other hand, notice that $V(E_{12}) - \{a, v, \ell', \ell''\}$ where $\ell' \in \{x, y, z\}$ and $\ell'' \in \{r, s, t\}$ is bipartite. Thus, we have shown that if $|V(M) \cap \{a, b, c\}| = 2$, then M has order 8.

Next, note that a similar argument could be used to show if $|V(M) \cap \{x, y, w\}| = 2$, then M has order 8. Thus, the last case to consider is when $|V(M) \cap \{a, b, c\}| \leq 1$ and $|V(M) \cap \{x, y, z\}| \leq 1$. But if one takes E_{12} and removes two vertices from each triangle, then the result is a subgraph of the 3-cube and is indeed bipartite. Thus, M has order 8. QED

Theorem 13 *The well-hued cubic graphs are $\{K_4, K_3 \square K_2, C_5 \square K_2, K_{3,3}, E_{12}\} \cup \mathcal{W}$.*

Proof. We have shown above that these graphs are the only candidates. It is clear that K_4 is well-hued. The remaining graphs are 3-colorable; thus, it only remains to check that they are well-bicovered. This is trivial for $K_{3,3}$. Graph $K_3 \square K_2$ is well-bicovered since any maximal bipartite subgraph contains exactly two vertices from each triangle. Further, in $C_5 \square K_2$ any maximal bipartite subgraph will contain exactly 4 vertices from each 5-cycle. Thus, $C_5 \square K_2$ is well-bicovered.

Finally, assume G is in $\mathcal{W}$. We claim that every maximal bipartite subgraph H consists of a maximal bipartite subgraph of each fragment—that is, has 4 vertices from each A and C and 6 vertices from each B —and thus has the same order. For suppose that H does not contain a maximum bipartite subgraph from some fragment say F . That is, there is a vertex $f \in F$ whose addition creates an odd cycle but not within F . It follows that F must be either an A or a B fragment. Further, in H there is a path from a neighbor on one side of the fragment F to a neighbor on the other side of the fragment. By inverting the second half of the (picture of) the fragment F if necessary, we may assume that the top left and the top right vertices of F have neighbors in H that are connected in H . Then one can easily verify that, regardless of whether these two neighbors are in the same partite set or not, one can always add a full maximal bipartite subgraph of the fragment F . In other words, if there were such a vertex f , then there would be some other vertex of F that one can add while preserving bipartiteness, a contradiction of the maximality of H . QED

5 Planar Graphs

In this section we consider some planar graphs.

The maximal outerplanar graphs that are well-bicovered were characterized in [8]. As an immediate consequence we obtain the characterization for well-hued graphs. We define a *whirlygig* as constructed as follows. Take a maximal outerplanar graph M with $m \geq 3$ vertices and let C represent the outer cycle of M . For each edge $u_i u_{i+1}$ on C , add a new vertex t_i adjacent only to u_i and u_{i+1} , and then add another vertex s_i that is adjacent to only t_i and u_i .

Theorem 14 *Maximal outerplanar well-hued graphs are the whirlygigs or K_3 .*

Proof. Since outerplanar graphs are 3-colorable, one need only check whether the graphs from [8] are well-covered or not. And they are except for the Hajòs graph. To see that a whirlygig is well-covered, note that the closed neighborhoods of the s_i are disjoint cliques and partition the vertex set; thus every maximal independent set has m vertices. QED

We consider next maximal planar well-hued graphs, also known as planar triangulations. In [5] Finbow et al. proved there are exactly four 4-connected well-covered planar triangulations, shown in Figure 6. Which of these are well-hued?

Theorem 15 *The graphs R_6 and R_8 are well-hued while R_7 and R_{12} are not.*

Proof. Note that $R_6 = K_{2,2,2}$, also known as the octahedron. A complete multipartite graph with equal-size parts is clearly well-hued.

Next, we show that R_8 is well-hued. Let A denote the degree-4 vertices (shown in white) and B the degree-5 vertices (shown in black). Note that the graph is symmetric; the orbits of the vertices are just A and B . Now, note that $\alpha_1(R_8) = 2$. (We know it is well-covered so just observe that two non-adjacent vertices of B are maximal.) Second note that $\chi(R_8) = 4$. (Four colors are sufficient since both A and B induce bipartite subgraphs; four colors are necessary since each color can only be used twice.) It also follows that $\alpha_2(R_8) \leq 4$ and $\alpha_3(R_8) \leq 6$.

Consider a maximal bipartite subgraph M of R_8 . If M has no vertex of B then it contains all of A and vice versa. It can be checked that if M contains one vertex of B then it contains three vertices of A ; if M contains two vertices of B then it contains two vertices of A ; and if it contains three vertices of B then it contains one vertex of A . Thus, R_8 is well-bicovered.

Consider a maximal 3-colorable subgraph S of R_8 . Again a case study is needed. If $|V(S) \cap B| = 4$, then S contains two vertices of A ; if $|V(S) \cap B| = 3$, then S contains three vertices of A ; and since removing any two vertices of B creates a 3-colorable graph, if $|V(S) \cap B| \leq 2$ then S contains all vertices of A and two vertices of B . Having exhausted all possibilities, we may conclude that every maximal 3-colorable subgraph has order 6. Thus, R_8 is well-hued.

To see that R_7 is not well-hued, note that $\{u_1, u_3, u_4, u_6, u_7, u_5\}$ and $\{u_1, u_2, u_3, u_4, u_5, u_6\}$ each induce a maximal 3-colorable subgraph. To see that R_{12} is not well-hued, note that $\{v_2, v_9, v_8, v_5, b, c\}$ and $\{v_1, v_2, v_3, v_5, v_4\}$ each induce a maximal bipartite subgraph. QED

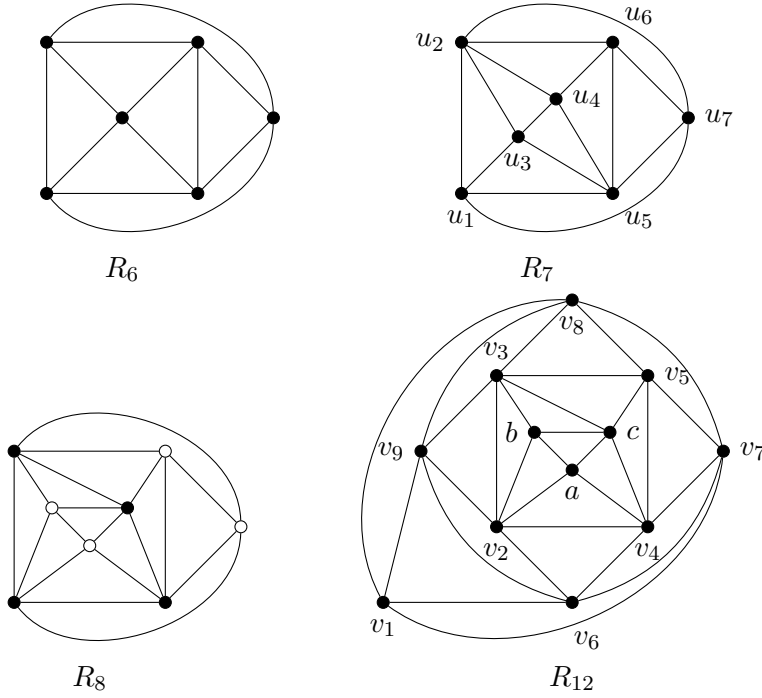


Figure 6: The 4-connected well-covered planar triangulations

In [6], Finbow et al. finished the characterization of well-covered plane triangulations by characterizing the infinite family of those well-covered triangulations that are exactly 3-connected. First, they constructed the following family. Take a planar graph where each component is a K_4 with at most one face non-empty, and arbitrarily add edges so as to form a connected planar graph where each face is a triangle. They denoted the resulting family of planar triangulations by $\mathcal{K}$.

Then they defined the following operation. Let F and G be graphs and suppose $abca$ and $rstr$ are triangular faces in F and G , respectively. Then an **O-join** of F and G at faces $abca$ and $rstr$ is a new graph H such that $V(H) = V(F) \cup V(G)$ and $E(H) = E(F) \cup E(G) \cup \{ar, as, bs, bt, ct, cr\}$. If H is the O-join of graphs F and G and the faces at which the O-join is made are irrelevant, we denote H by $F \circ G$. If G is a planar well-covered graph and $abca$ is a triangular face of G , then $abca$ is called a **YES-face** if each of $G - a - b$, $G - b - c$, and $G - a - c$ is also well-covered.

Finally, they defined the family $\mathcal{K}'$ as the collection of all graphs that can be obtained from a member of G of $\mathcal{K}$ by choosing two disjoint sets R and S (either or

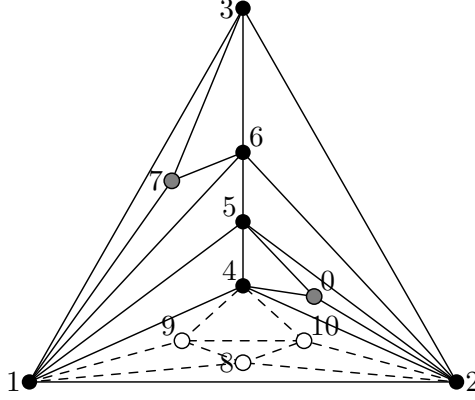


Figure 7: A graph in $\mathcal{K}'$ that is not well-hued

both possibly empty) of YES-faces in G and O-joining a triangle to each face in R , and O-joining a copy of R_8 to each face of S via a YES-face of R_8 .

Theorem 16 [6] *Let G be a planar triangulation. Then G is well-covered if and only if G is in $\mathcal{K}'$ or one of eight exceptional graphs.*

It is unclear what happens for well-hued triangulations. It is true that if the O-join produces a well-hued graph, then the original must be well-hued too, as we now show.

Lemma 9 *If $G \odot K_3$ is well-hued, then so is G .*

Proof. Every maximal i -partite set of G can be extended to a maximal i -partite set of $G \odot K_3$ containing exactly $\min(i, 3)$ vertices of the new triangle. Therefore, if G is not well- k -partite then neither is $G \odot K_3$. QED

Unfortunately, there are examples of graphs where the O-join preserves well-huedness and where it does not. For example, the graph in Figure 7 is in $\mathcal{K}'$: it is formed by the O-join of a triangle in face bounded by vertices labeled 1, 2, 4. But both the set $\{0, 1, 2, 7, 9, 10\}$ and the set $\{0, 1, 6, 9, 10\}$ induce a maximal bipartite subgraph. Also, there are graphs in the base family $\mathcal{K}$ that are well-hued and graphs that are not: for example, of the 48 elements of $\mathcal{K}$ of order 12, exactly 13 are well-hued.

6 One Further Thought

Apart from the open problems mentioned in the text, the question of the computational complexity remains open. Sankaranarayana, and Stewart [14] showed that a straight-forward reduction from SAT is enough to show that testing for a graph to be well-covered is co-NP-complete. But attempts to adapt the reduction for well-bicovered or well-hued have been unsuccessful.

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